

MAINTENANCE OPERATIONS CLIENT

Data Dictionary

Maintenance Operations Analytics & Reporting Automation

Document Control	Detail
Engagement	Aug-Nov 2024
Role	Business Analyst + Data Analyst
Primary Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Confirmed Sources	Excel, CSV, Google Sheets and database
Related Artifact	KPI Catalogue
Status	Portfolio reconstruction grounded in confirmed field-level evidence

Integrity note: Only fields remembered from the engagement are represented as confirmed. Exact original database column names, source-table names, SQL data types, field lengths, nullability constraints, keys and coded value lists are not retained and are not fabricated.

1. Purpose

This Data Dictionary defines the business meaning and analytical role of the confirmed maintenance fields used in the case study. It provides a controlled vocabulary between source preparation, KPI definition, Power BI reporting and testing.

Design principle: The dictionary describes business concepts first. Technical implementation details are added only as portfolio-reconstructed specifications when they are needed to rebuild the analytical solution.

2. Data Landscape

Attribute	Confirmed Information
Data history	Historical maintenance data
Approximate volume	10,000 records
Operational coverage	20 teams
Source formats	Excel, CSV, Google Sheets and database
Record domains	Maintenance tasks/work orders, inspections, service requests and equipment-related records
Known quality issues	Duplicates, missing values, inconsistent team names, inconsistent date formats, inconsistent status names and blank records

3. Field Classification

Classification	Meaning
Confirmed field	The field/business concept is remembered from the engagement.
Reconstructed analytical type	A practical portfolio data type used to rebuild the model; not claimed as the original database type.
Reconstructed validation rule	A defensible data-quality rule for portfolio implementation; not claimed as the exact historical ETL rule.
Not retained	Original schema metadata or coded value definitions are unavailable.

4. Core Data Dictionary

Field	Domain	Portfolio Type*	Business Definition	Analytical Use	Quality / Governance Rule
Task / Work Order ID	Identifier	Text / identifier	Identifies a maintenance task or work-order record.	Counts, record-level validation, status analysis, completion, overdue, backlog	Must be checked for duplicates and blanks; exact original uniqueness/key constraint is not retained.
Team	Organization	Text / category	Identifies the operational team associated with a maintenance record.	Team-wise completion, workload comparison, performance analysis	Standardize inconsistent team names. Exact historical team-name mapping is not retained.
Technician	Resource	Text / category	Identifies the technician associated with maintenance activity.	Technician-level workload and performance analysis	Preserve missing/unknown attribution transparently; do not invent technician assignment.
Asset / Equipment	Asset	Text / category	Identifies the asset or equipment associated with maintenance activity.	Asset/equipment breakdown, recurring activity and issue-pattern analysis	Standardize labels where required. Do not infer failure causation from record frequency alone.
Created Date	Date	Date / datetime	Represents the creation/initiation date available for the maintenance record.	Historical trend context; possible reconstructed completion-time calculation	Standardize date format and validate parseability. Exact historical use as completion-time start point is not retained.
Due Date	Date	Date / datetime	Represents the date by which maintenance work is due.	Overdue classification and due-work analysis	Must be a valid date for overdue classification.
Completion Date	Date	Date / datetime	Represents the recorded completion date for completed maintenance work.	Completion-time analysis and completion chronology	Validate parseability and chronology; do not fabricate missing dates.
Status	Workflow	Text / category	Represents the maintenance record's operational status.	Completed, open, pending, overdue and backlog-related reporting	Standardize inconsistent status names. Exact original status-value list/mapping is not retained.
Priority	Criticality	Text / category	Represents the assigned priority/criticality of maintenance work.	Priority breakdown and high-priority pending analysis	Standardize supported values; preserve unknown/missing values rather than inventing priority.
Category	Classification	Text / category	Represents the maintenance category/type associated with the record.	Recurring-category analysis and operational segmentation	Standardize inconsistent labels before aggregation; exact original category taxonomy is not retained.
Location	Operational context	Text / category	Represents the location associated with maintenance activity.	Location-based slicing where analytically appropriate	Use only where valid and non-sensitive; exact location hierarchy is not retained.
Remarks	Context	Text / free text	Contains supporting remarks or operational notes associated with the record.	Contextual investigation and qualitative support where appropriate	Treat as unstructured text; avoid exposing sensitive operational or personal information.

*** Portfolio Type:** The analytical types above are reconstruction choices for a portfolio model. They are not claimed as the original SQL/database field types.

5. Field-Level KPI Traceability

Field	KPI / Analysis Dependencies
Task / Work Order ID	Assigned work, completed/open/pending counts, overdue, backlog, priority/category/asset aggregations
Team	Team-wise completion, workload imbalance, team performance
Technician	Technician performance, technician workload
Asset / Equipment	Asset/equipment breakdown and issue-pattern analysis
Created Date	Trend analysis; reconstructed completion-time logic if selected
Due Date	Confirmed overdue rule
Completion Date	Completion-time analysis
Status	Completion Rate, completed/open/pending, overdue, backlog reconstruction
Priority	Priority breakdown, high-priority pending work
Category	Recurring maintenance-category analysis
Location	Location-based operational analysis where used
Remarks	Contextual investigation; not a primary quantitative KPI input

6. Confirmed Business-Rule Dependencies

Completion Rate: Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100

Required Concept	Field Dependency / Control
Completed Maintenance Tasks	Status must support a documented Completed classification.
Total Assigned Maintenance Tasks	Eligible maintenance population must be defined consistently in the portfolio rebuild.
Record counting	Task / Work Order ID and duplicate controls must prevent inflated counts.

Overdue: Due Date < Current Date AND Status != Completed

Required Concept	Field Dependency / Control
Due Date	Must be valid and earlier than Current Date.
Status	Must not be classified as Completed.
Missing Due Date	Must not be falsely classified as overdue.
Record identity	Task / Work Order ID supports record-level exception review.

7. Status Field Governance

Status is a critical analytical field because it influences completion, open work, pending work, overdue reporting and the reconstructed backlog logic.

Governance Requirement	Rule
Standardization	Normalize inconsistent source status labels to documented analytical categories.
Completed	A controlled Completed category is required for the confirmed completion and overdue rules.
Open vs Pending	Do not assume Open and Pending are equivalent unless the rebuilt mapping explicitly establishes that meaning.
Unknown status	Retain as unknown/unmapped or handle through a documented rule; do not silently force into a valid status.
Historical mapping boundary	Exact original source values and mapping table are not retained.

8. Date Field Governance

Field	Validation / Analytical Control
Created Date	Parse to a valid date/datetime. Use in completion-time calculations only if the portfolio reconstruction explicitly selects it as the start/reference date.
Due Date	Parse to a valid date/datetime. Required for the confirmed overdue rule.
Completion Date	Parse to a valid date/datetime. Use for completed-work timing analysis where valid.
Chronology	Investigate impossible/negative intervals rather than silently accepting them.
Missing dates	Do not invent dates. Exclude or classify records according to the documented KPI rule.
Format inconsistency	Standardize inconsistent source formats before calculations.

9. Organizational & Resource Field Governance

Field	Governance Rule
Team	Normalize spelling/label variants into a controlled reporting representation.
Technician	Preserve actual attribution where available; missing values remain explicit.
Location	Use only at the level supported by source data and confidentiality constraints.
20 teams	The project scope is approximately 10,000 historical records across 20 teams; do not invent the names or exact organizational classification of those teams.

10. Asset, Priority & Category Governance

Field	Governance Rule
Asset / Equipment	Use the available identifier/label for grouping; do not invent asset hierarchy or equipment metadata.
Priority	Standardize supported source values; exact original priority vocabulary is not retained.
Category	Standardize labels sufficiently for recurring-category analysis; exact historical taxonomy is not retained.
Interpretation	High frequency or backlog by category/asset is an analytical signal and does not by itself establish root cause.

11. Missing-Value Treatment Framework

Field Type	Portfolio Handling Principle
Identifiers	Investigate blanks/duplicates because they affect reliable record counting.
Team / Technician	Retain unknown/missing attribution transparently unless an evidence-based correction exists.
Dates	Do not impute unsupported operational dates.
Status	Do not fabricate a status; use a documented unknown/unmapped treatment.
Priority / Category	Preserve missing/unknown values unless a source-supported mapping exists.
Remarks	Blank remarks do not require synthetic text.

12. Duplicate-Control Framework

Duplicates were a confirmed source-data issue. The exact historical duplicate-detection algorithm is not retained, so the portfolio implementation must define its rule explicitly.

Control Step	Portfolio Requirement
Detection	Identify exact and/or candidate duplicate records using documented fields.
Review	Distinguish true duplicate rows from legitimately repeated maintenance activity.
Resolution	Remove, consolidate or retain records according to a documented evidence-based rule.
KPI protection	Ensure duplicate handling occurs before counts/rates that could otherwise be inflated.
Auditability	Document the reconstructed rule and validation result.

Do not assume: Task / Work Order ID is analytically useful as an identifier, but the exact historical primary-key constraint is not retained.

13. Reconstructed Analytical Schema Boundary

A later portfolio Data Model & DAX document may organize these fields into fact and dimension structures for a clean Power BI model. That model will be a portfolio reconstruction. This dictionary does not claim that the historical solution used a particular star schema, table name, relationship cardinality or database design.

Not Retained	Portfolio Treatment
Original table names	Use clearly labeled reconstructed table names if needed.
Original column names	Use the business-friendly field names in this dictionary unless evidence supports another name.
SQL data types	Use reconstructed analytical types and label them accordingly.
Primary / foreign keys	Design portfolio relationships without claiming historical equivalence.
Nullability / field length	Define only if required for the rebuild; do not present as original schema metadata.
Original status/priority/category codes	Create sanitized mapping values only as portfolio reconstruction.

14. Data Quality Validation Checklist

- Check record counts before and after cleansing.
- Identify and control duplicate records.
- Profile missing values in analytically important fields.
- Standardize team labels before team-level reporting.
- Standardize status labels before completion/open/pending/overdue calculations.
- Parse and validate Created Date, Due Date and Completion Date.
- Preserve unknown priority/category values unless a supported mapping exists.
- Verify Completion Rate against its confirmed numerator and denominator definition.
- Verify every overdue record satisfies the confirmed due-date/status rule.
- Confirm no confidential client or sensitive operational data appears in the public portfolio.

15. Traceability to Project Artifacts

Artifact	How This Dictionary Supports It
BRD	Provides the field vocabulary required by data, analytical and reporting requirements.
User Stories & Acceptance Criteria	Defines the fields needed to make story acceptance criteria testable.
KPI Catalogue	Provides field dependencies and quality controls for governed KPI definitions.
Data Cleaning Methodology	Defines what must be standardized, validated or controlled.
Operational Analysis	Provides consistent dimensions for team, technician, category, priority and asset analysis.
Power BI Requirements	Defines business-friendly fields available for reporting/filtering.
Data Model & DAX	Provides the controlled semantic layer for reconstructed model design.
Dashboard Testing	Provides field-level validation points for KPI reconciliation.

16. Confidentiality & Portfolio Sanitization

Client identity: The real client name is confidential. Public artifacts use Maintenance Operations Client.

- Do not publish real technician names or other employee-identifying information.
- Do not publish confidential locations, asset identifiers or operational remarks.
- Do not invent the names of the 20 teams.
- Use sanitized or synthetic values in reconstructed dashboard/data examples.
- Clearly label reconstructed technical metadata and example values.
- Retain the confirmed 92% completion rate only with its correct interpretation as an observed KPI.